**M467 Assigned Pins**:

| Name | Signal |
|---|---|
| PB.1 | EADC0_CH0 |
| PB.0 | EADC0_CH1 |
| PC.13 | CLKO |
| PD.12 | CLKO |
| PF.5 | X32_IN |
| PF.4 | X32_OUT |
| PF.3 | XT1_IN |
| PF.2 | XT1_OUT |
| PE.8 | RMII_MDC |
| PE.9 | RMII_MDIO |
| PE.10 | RMII_TXD0 |
| PE.11 | RMII_TXD1 |
| PE.12 | RMII_TXDEN |
| PC.8 | RMII_REFCLK |
| PC.7 | RMII_RXD0 |
| PC.6 | RMII_RXD1 |
| PA.7 | RMII_RXDV |
| PA.6 | RMII_RXERR |
| PF.0 | SWD_DAT |
| PF.1 | SWD_CLK |
| PC.5 | UART2_TX |
| PC.4 | UART2_RX |
| PD.5 | L.DIR |
| PD.4 | L.EN |
| PD.3 | L.STO |
| PD.2 | R.DIR |
| PD.1 | R.EN |
| PD.0 | R.STO |
| PA.13 | USB_D_N |
| PA.14 | USB_D_P |
| PE.7 | CAN1_TX |
| PE.6 | CAN1_RX |
| PH.10 | UART0_TX |
| PH.11 | UART0_RX |
| PJ.10 | CAN0_TX |
| PJ.11 | CAN0_RX |
| PJ.13 | BPWM0_CH0 |
| PB.13 | DAC0_OUT |
| PB.12 | DAC1_OUT |
| PB.11 | BPWM1_CH0 |

**Hardware layout and configurations:**

**A. Connector**

| Connector | Pin | M467 pin | Function |
|---|---|---|---|

| | | | | | | |
|---|---|---|---|---|---|---|
| J1<br>Power Supply | 1 | | - | | 12V INPUT | |
| | 2 | | - | | GND | |
| J2<br>USB | 1 | | - | | VBUS | |
| | 2 | | PA.13 | | USB_D_N | |
| | 3 | | PA.14 | | USB_D_P | |
| | 4 | | - | | - | |
| | 5 | | - | | GND | |
| | 6 | | - | | Shield | |
| J3<br>NULINK | 1 | | - | | 3V3 | |
| | 2 | | PF.0 | | SWD_DAT | |
| | 3 | | PF.1 | | SWCLK | |
| | 4 | | NRST | | NRST | |
| | 5 | | - | | GND | |
| J4<br>Left M467<br>Extension Pins | 1 | 2 | - | - | GND | 3V3 |
| | 3 | 4 | - | - | GND | 3V3 |
| | 5 | 6 | PB.5 | PB.4 | PB.5 | PB.4 |
| | 7 | 8 | PB.3 | PB.2 | PB.3 | PB.2 |
| | 9 | 10 | PC.12 | PC.11 | PC.12 | PC.11 |
| | 11 | 12 | PC.10 | PC.9 | PC.10 | PC.9 |
| | 13 | 14 | PB.1 | PB.0 | PB.1 | PB.0 |
| | 15 | 16 | PA.11 | PA.10 | PA.11 | PA.10 |
| | 17 | 18 | PA.9 | PA.8 | PA.9 | PA.8 |
| | 19 | 20 | PC.13 | PD.12 | PC.13 | PD.12 |
| | 21 | 22 | PD.10 | PD.11 | PD.10 | PD.11 |
| | 23 | 24 | PG.1 | PG.0 | PG.1 | PG.0 |
| | 25 | 26 | PG.3 | PG.2 | PG.3 | PG.2 |
| | 27 | 28 | PI.6 | PG.4 | PI.6 | PG.4 |
| | 29 | 30 | PI.8 | PI.7 | PI.8 | PI.7 |
| | 31 | 32 | PI.10 | PI.9 | PI.10 | PI.9 |
| | 33 | 34 | PF.11 | PI.11 | PF.11 | PI.11 |
| | 35 | 36 | PF.9 | PF.10 | PF.9 | PF.10 |
| | 37 | 38 | PF.7 | PF.8 | PF.7 | PF.8 |
| | 39 | 40 | PF.5 | PF.6 | PF.5 | PF.6 |
| J5<br>Bottom M467<br>Extension Pins | 1 | 2 | PF.4 | PH.0 | PF.4 | PH.0 |
| | 3 | 4 | PH.1 | PH.2 | PH.1 | PH.2 |
| | 5 | 6 | PH.3 | PH.4 | PH.3 | PH.4 |
| | 7 | 8 | PH.5 | PH.6 | PH.5 | PH.6 |
| | 9 | 10 | PH.7 | PF.3 | PH.7 | PF.3 |
| | 11 | 12 | PF.2 | PE.8 | PF.2 | PE.8 |
| | 13 | 14 | PE.9 | PE.10 | PE.9 | PE.10 |
| | 15 | 16 | PE.11 | PE.12 | PE.11 | PE.12 |
| | 17 | 18 | PE.13 | PC.8 | PE.13 | PC.8 |
| | 19 | 20 | PC.7 | PC.6 | PC.7 | PC.6 |
| | 21 | 22 | PA.6 | PA.7 | PA.6 | PA.7 |
| | 23 | 24 | PI.13 | PI.12 | PI.13 | PI.12 |
| | 25 | 26 | PI.15 | PI.14 | PI.15 | PI.14 |

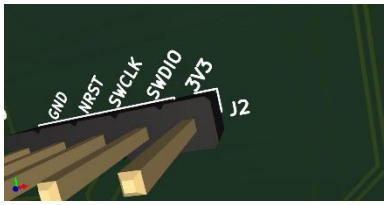
| | | | | | | |
|---|---|---|---|---|---|---|
| | 27 | 28 | PJ.1 | PJ.0 | PJ.1 | PJ.0 |
| | 29 | 30 | PA.4 | PA.5 | PA.4 | PA.5 |
| | 31 | 32 | PA.2 | PA.3 | PA.2 | PA.3 |
| | 33 | 34 | PA.0 | PA.1 | PA.0 | PA.1 |
| | 35 | 36 | PE.15 | PE.14 | PE.15 | PE.14 |
| | 37 | 38 | PF.1 | PF.0 | PF.1 | PF.0 |
| | 39 | 40 | PD.8 | PD.9 | PD.8 | PD.9 |
| J6<br>Right M467<br>Extension Pins | 1 | 2 | PC.5 | PC.4 | PC.5 | PC.4 |
| | 3 | 4 | PC.3 | [C.2 | PC.3 | [C.2 |
| | 5 | 6 | PC.1 | PC.0 | PC.1 | PC.0 |
| | 7 | 8 | PG.9 | PG.10 | PG.9 | PG.10 |
| | 9 | 10 | PG.11 | PG.12 | PG.11 | PG.12 |
| | 11 | 12 | PG.13 | PG.14 | PG.13 | PG.14 |
| | 13 | 14 | PG.15 | PJ.2 | PG.15 | PJ.2 |
| | 15 | 16 | PG.3 | PJ.4 | PG.3 | PJ.4 |
| | 17 | 18 | PJ.5 | PJ.6 | PJ.5 | PJ.6 |
| | 19 | 20 | PJ.7 | PH.12 | PJ.7 | PH.12 |
| | 21 | 22 | PH.14 | PH.13 | PH.14 | PH.13 |
| | 23 | 24 | PD.7 | PH.15 | PD.7 | PH.15 |
| | 25 | 26 | PD.5 | PD.6 | PD.5 | PD.6 |
| | 27 | 28 | PD.3 | PD.4 | PD.3 | PD.4 |
| | 29 | 30 | PD.1 | PD.2 | PD.1 | PD.2 |
| | 31 | 32 | PD.13 | PD.0 | PD.13 | PD.0 |
| | 33 | 34 | PA.13 | PA.12 | PA.13 | PA.12 |
| | 35 | 36 | PA.15 | PA.14 | PA.15 | PA.14 |
| | 37 | 38 | - | - | GND | 3V3 |
| | 39 | 40 | - | - | GND | 3V3 |
| J7<br>Top M467<br>Extension Pins | 1 | 2 | - | - | GND | 3V3 |
| | 3 | 4 | PE.7 | PE.6 | PE.7 | PE.6 |
| | 5 | 6 | PE.5 | PE.4 | PE.5 | PE.4 |
| | 7 | 8 | PE.3 | PE.2 | PE.3 | PE.2 |
| | 9 | 10 | PE.1 | PE.0 | PE.1 | PE.0 |
| | 11 | 12 | PH.8 | PH.9 | PH.8 | PH.9 |
| | 13 | 14 | PH.10 | PH.11 | PH.10 | PH.11 |
| | 15 | 16 | PD.14 | PJ.8 | PD.14 | PJ.8 |
| | 17 | 18 | PJ.9 | PJ.10 | PJ.9 | PJ.10 |
| | 19 | 20 | PJ.11 | PJ.12 | PJ.11 | PJ.12 |
| | 21 | 22 | PG.5 | PJ.13 | PG.5 | PJ.13 |
| | 23 | 24 | PG.7 | PG.6 | PG.7 | PG.6 |
| | 25 | 26 | PC.14 | PG.8 | PC.14 | PG.8 |
| | 27 | 28 | PB.14 | PB.15 | PB.14 | PB.15 |
| | 29 | 30 | PB.12 | PB.13 | PB.12 | PB.13 |
| | 31 | 32 | PB.10 | PB.11 | PB.10 | PB.11 |
| | 33 | 34 | PB.8 | PB.9 | PB.8 | PB.9 |
| | 35 | 36 | PB.6 | PB.7 | PB.6 | PB.7 |
| | 37 | 38 | - | - | GND | 3V3 |

| | 39 | 40 | - | - | GND | 3V3 |
|---|---|---|---|---|---|---|
| J8<br>RJ45 | 1 | | - | | TX+ | |
| | 2 | | - | | TX- | |
| | 3 | | - | | RX+ | |
| | 4 | | - | | - | |
| | 5 | | - | | - | |
| | 6 | | - | | RX- | |
| | 7 | | - | | - | |
| | 8 | | - | | - | |
| | 9 | | - | | LED0- | |
| | 10 | | - | | LED0+ | |
| | 11 | | - | | LED1- | |
| | 12 | | - | | LED1+ | |
| | SH | | - | | Shield (connected to Earth) | |
| J9<br>RELAY_R.STO | 1 | | - | | NO1 | |
| | 2 | | - | | COM1 | |
| | 3 | | - | | NC1 | |
| J10<br>RELAY_R.EN | 1 | | - | | NO2 | |
| | 2 | | - | | COM2 | |
| | 3 | | - | | NC2 | |
| J11<br>RELAY_R.DIR | 1 | | - | | NO3 | |
| | 2 | | - | | COM3 | |
| | 3 | | - | | NC3 | |
| J12<br>RELAY_L.STO | 1 | | - | | NO4 | |
| | 2 | | - | | COM4 | |
| | 3 | | - | | NC4 | |
| J13<br>RELAY_L.EN | 1 | | - | | NO5 | |
| | 2 | | - | | COM5 | |
| | 3 | | - | | NC5 | |
| J14<br>RELAY_L.DIR | 1 | | - | | NO6 | |
| | 2 | | - | | COM6 | |
| | 3 | | - | | NC6 | |
| J15<br>BPWM0 | 1 | | PJ.13 | | BPWM0 SIGNAL | |
| | 2 | | - | | GND | |
| J16<br>BPWM1 | 1 | | PB.11 | | BPWM1 SIGNAL | |
| | 2 | | - | | GND | |
| J17<br>OUTPUT_DAC0 | 1 | | PB.12 | | OUTPUT_DAC | |
| | 2 | | - | | GND | |
| J18<br>OUTPUT_DAC1 | 1 | | PB.13 | | OUPUT_DAC | |
| | 2 | | - | | GND | |
| J19<br>EADC0_CH1 | 1 | | - | | 5V | |
| | 2 | | PB.1 | | SIGNAL | |
| | 3 | | - | | GND | |
| J20<br>EADC0_CH0 | 1 | | - | | 5V | |
| | 2 | | PB.0 | | SIGNAL | |
| | 3 | | - | | GND | |

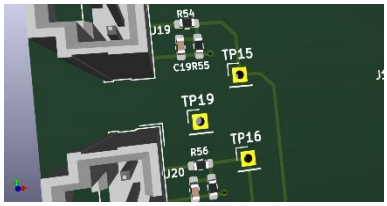
| | | | |
|---|---|---|---|
| J21<br>CAN0 TEST | 1 | - | CAN0L |
| | 2 | - | CAN0H |
| J22<br>CAN1 TEST | 1 | - | CAN1L |
| | 2 | - | CAN1H |
| J23<br>CAN0 | 1 | - | CAN0L |
| | 2 | - | CAN0H |
| | 3 | PJ.11 | CAN0_RX |
| | 4 | PJ.10 | CAN0_TX |
| | 5 | - | GND |
| | 6 | - | 5V |
| J24<br>CAN1 | 1 | - | CAN1L |
| | 2 | - | CAN1H |
| | 3 | PE.6 | CAN1_RX |
| | 4 | PE.7 | CAN1_TX |
| | 5 | - | GND |
| | 6 | - | 5V |
| J25<br>UART0 | 1 | - | GND |
| | 2 | PH.11 | UART0_RX |
| | 3 | PH.10 | UART0_TX |
| J26<br>UART2 | 1 | - | GND |
| | 2 | PC.4 | UART2_RX |
| | 3 | PC.5 | UART2_TX |

1st pin indicators:

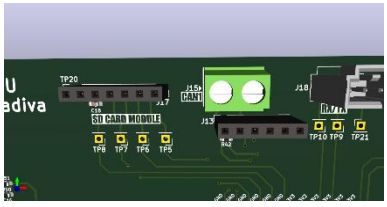
- Pin headers: Right-angle mark at the side of the connector.



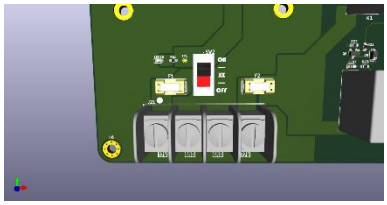

- Molex connectors: Right-angle mark at the side of the connector.




- Small terminal block: White triangle mark next to the connector.




- Power supply terminal block: White circle towards the first pin of the connector.



**B. Switch**

S1 – Power switch
S2 – Reset switch

**C. LED**

LED1 – 5V power
LED2 – 3V3 power
LED3 – 12V power
LED4 – VBUS power
LED5 – R.STO
LED6 – R.EN
LED7 – R.DIR
LED8 – L.STO
LED9 – L.EN
LED10 – L.DIR

**D. Test Point**

TP1 – 12V power
TP2 – 5V power
TP3 – 3V3 power
TP4, TP5, TP6, TP7, TP8 – Ground point for test points (used to give a closer ground point to an area of test points)
TP9, TP10 – Microcontroller clock output
TP11 – BPWM0 SIGNAL
TP12 – BPWM1 SIGNAL
TP13 – DAC0_OUT
TP14 – DAC1_OUT
TP15 – EADC0_CH1
TP16 – EADC0_CH0
TP17 – CAN0_RX
TP18 – CAN0_TX
TP19 – CAN1_RX
TP20 – CAN1_TX
TP21 – UART0_RX
TP22 – UART0_TX
TP23 – UART2_RX
TP24 – UART2_TX

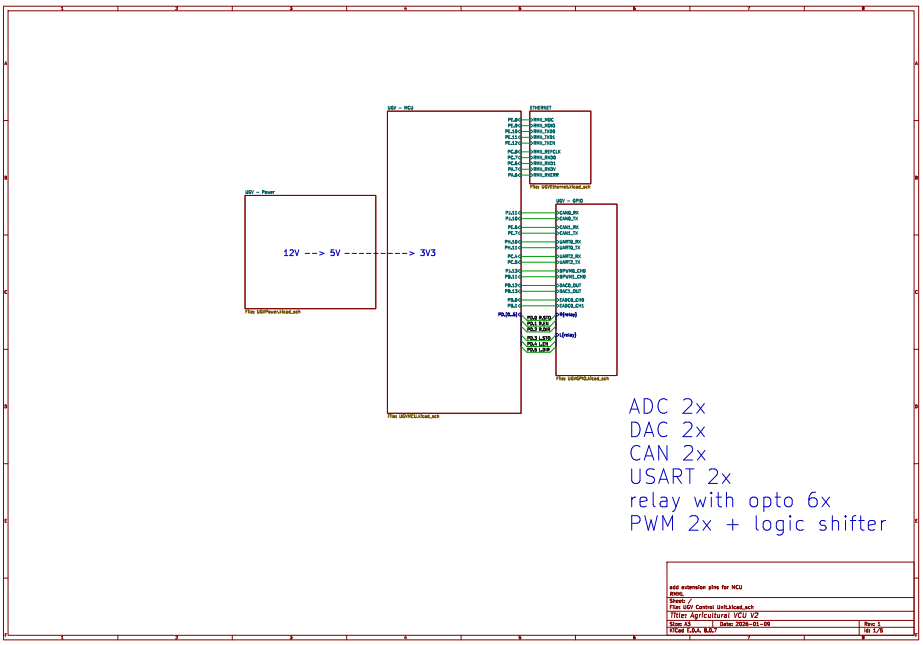
**E. PCB Standoff**

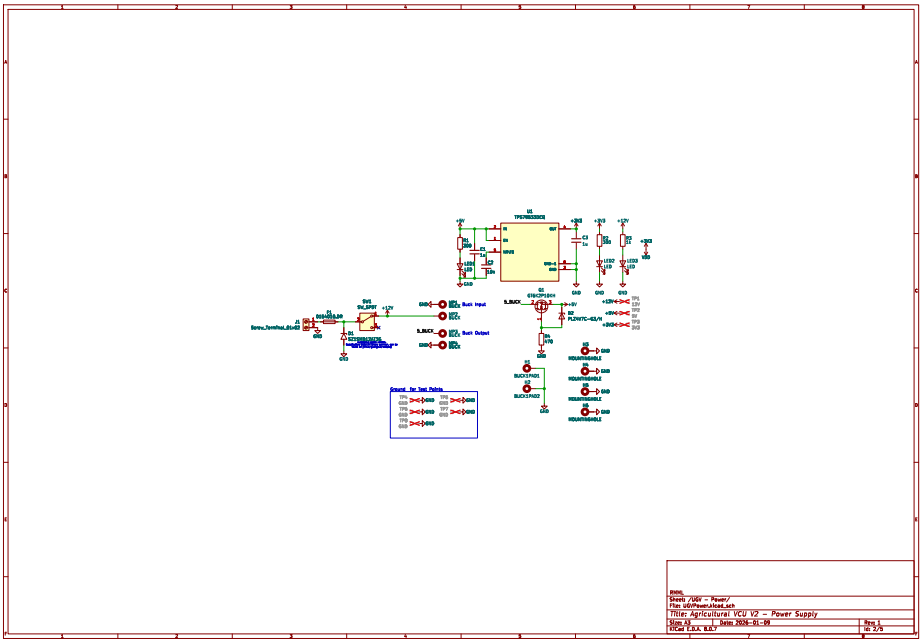
MP1 – GND
MP2 – BUCK INPUT
MP3 – BUCK OUTPUT
MP4 – GND

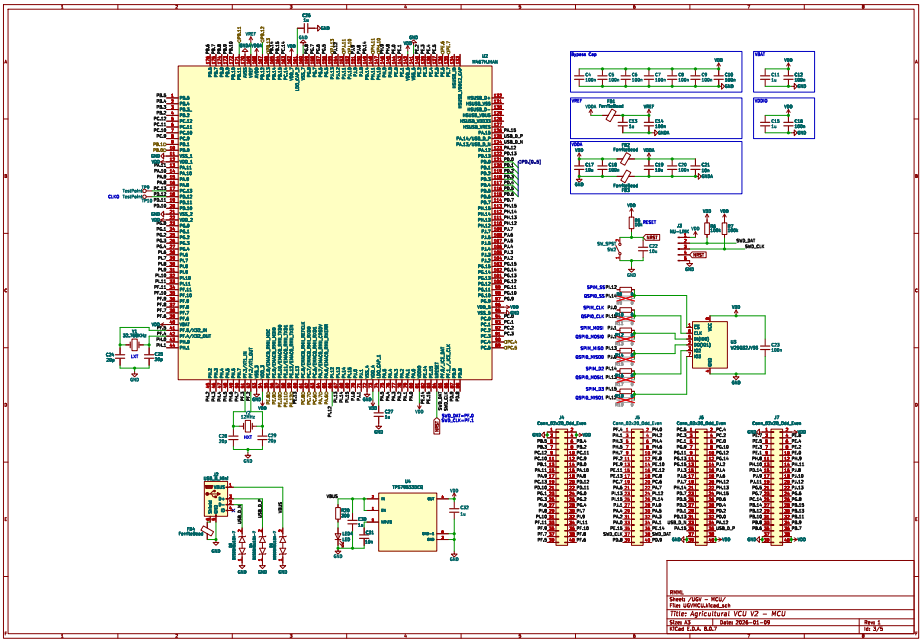
**F. Fuse**

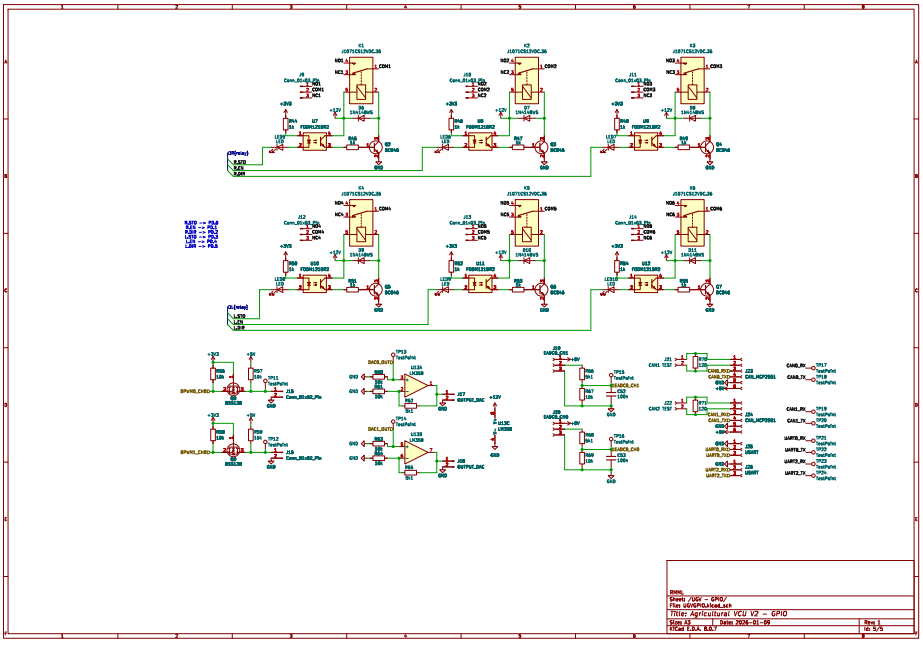
F1 – 12V input

## G. Schematic




Bypass Cap
VBAT
VDDA
VDD
VDD_MCU
VDD_3V3
VDD_5V
VDD_1V8
VDD_1V2
VDD_ETH
VDD_ETH_1V2
VDD_ETH_3V3
VDD_ETH_2V5
VDD_ETH_1V8
VDD_ETH_1V0
VDD_ETH_0V6
VDD_ETH_0V3
VDD_ETH_0V1
VDD_ETH_0V0
VDD_ETH_0V0_1
VDD_ETH_0V0_2
VDD_ETH_0V0_3
VDD_ETH_0V0_4
VDD_ETH_0V0_5
VDD_ETH_0V0_6
VDD_ETH_0V0_7
VDD_ETH_0V0_8
VDD_ETH_0V0_9
VDD_ETH_0V0_10
VDD_ETH_0V0_11
VDD_ETH_0V0_12
VDD_ETH_0V0_13
VDD_ETH_0V0_14
VDD_ETH_0V0_15
VDD_ETH_0V0_16
VDD_ETH_0V0_17
VDD_ETH_0V0_18
VDD_ETH_0V0_19
VDD_ETH_0V0_20
VDD_ETH_0V0_21
VDD_ETH_0V0_22
VDD_ETH_0V0_23
VDD_ETH_0V0_24
VDD_ETH_0V0_25
VDD_ETH_0V0_26
VDD_ETH_0V0_27
VDD_ETH_0V0_28
VDD_ETH_0V0_29
VDD_ETH_0V0_30
VDD_ETH_0V0_31
VDD_ETH_0V0_32
VDD_ETH_0V0_33
VDD_ETH_0V0_34
VDD_ETH_0V0_35
VDD_ETH_0V0_36
VDD_ETH_0V0_37
VDD_ETH_0V0_38
VDD_ETH_0V0_39
VDD_ETH_0V0_40




| | |
|---|---|
| Sheet: /GEV - GPIO/ | |
| File: GEVGPIO.kicad_sch | |
| Title: Agricultural VCU V2 - GPIO | |
| Size: A3 | Rev: 1 |
| KiCad E.D.A. 8.0.7 | Id: 5/5 |